Tier 1 Tier 2 Tier 3 Tier 4

(a) r = -0.12
RDF error [%]
eSEN-30M-OAM
GRACE-2L-OAM
MACE-MP-0
MACE-MPA-0
NequIP-OAM-L
MACE-MH-1-OMAT
EquiformerV2
MatterSim-v1.0.0-5M
UMA-S-P1
rb-v2
orb-v3-conservative-inf-omat
UMA-M-P1
orb-v3-direct-20-omat
PET-OMAT-XL
PET-OAM-XL

(b) r = -0.31
GRACE-2L-OAM
eSEN-30M-OAM
MACE-MP-0
MACE-MPA-0
NequIP-OAM-L
EquiformerV2
orb-v2
MatterSim-v1.0.0-5M
orb-v3-conservative-inf-omat
PET-OAM-XL

(c) r = -0.18
eSEN-30M-OAM
GRACE-2L-OAM
MACE-MP-0
NequIP-OAM-L
EquiformerV2
MACE-MPA-0
MatterSim-v1.0.0-5M
orb-v2
orb-v3-conservative-inf-omat
PET-OAM-XL

(d) r = 0.47
Pressure histogram error [%]
MACE-MPA-0
MACE-MP-0
orb-v3-direct-20-omat
EquiformerV2
MatterSim-v1.0.0-5M
orb-v3-conservative-inf-omat
NequIP-OAM-L
orb-v2
PET-OAM-XL
GRACE-2L-OAM
PET-OMAT-XL
UMA-S-P1
eSEN-30M-OAM
UMA-M-P1
MACE-MH-1-OMAT

(e) r = -0.68
MACE-MP-0
MACE-MPA-0
MatterSim-v1.0.0-5M
EquiformerV2
orb-v3-conservative-inf-omat
orb-v2
NequIP-OAM-L
PET-OAM-XL
GRACE-2L-OAM
eSEN-30M-OAM

(f) r = 0.27
MACE-MPA-0
MACE-MP-0
MatterSim-v1.0.0-5M
orb-v3-conservative-inf-omat
EquiformerV2
NequIP-OAM-L
orb-v2
PET-OAM-XL
GRACE-2L-OAM
eSEN-30M-OAM

(g) r = 0.10
VDOS error [%]
MACE-MP-0
eSEN-30M-OAM
GRACE-2L-OAM
NequIP-OAM-L
MACE-MPA-0
MACE-MH-1-OMAT
orb-v2
EquiformerV2
MatterSim-v1.0.0-5M
orb-v3-direct-20-omat
UMA-S-P1
UMA-M-P1
orb-v3-conservative-inf-omat
PET-OMAT-XL
PET-OAM-XL
Force RMSE [eV/Å]

(h) r = -0.56
MACE-MP-0
eSEN-30M-OAM
GRACE-2L-OAM
NequIP-OAM-L
MACE-MPA-0
MatterSim-v1.0.0-5M
orb-v2
EquiformerV2
orb-v3-conservative-inf-omat
PET-OAM-XL
F1 score

(i) r = -0.07
MACE-MP-0
eSEN-30M-OAM
GRACE-2L-OAM
MACE-MPA-0
NequIP-OAM-L
orb-v2
MatterSim-v1.0.0-5M
EquiformerV2
orb-v3-conservative-inf-omat
PET-OAM-XL
κ_{SRME} score